Deniz Erdogan

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Numerical Simulation · Automation · Machine Learning · Data Analysis

Education

2023 – 2025 (exp.) **M.Sc. in Computational Science and Engineering** München, Germany
Technische Universität München
- Current Grade: 1.80 (German System)
- Research on computational methods, simulations, machine learning, data science
- **Thesis**: Physics Enhanced Machine Learning: Learning Dynamic Systems using Hamiltonian Neural Networks

2018 – 2023 **B.Sc. in Computer Science and Mechanical Engineering** Istanbul, Turkey
Koç University
- GPA: 3.91/4.00
- High Honor Student with scholarship; highest ranking student in Mechanical Engineering
- **Thesis**: Manufacturing and Design of a Fully Automated TRV Chair for BPP Vertigo Diagnosis and Treatment

Work Experience

Nov 2024 – Present **B.B.W Automotive Consulting** München, Germany
Software Engineer. Working Student
- Developed quality-inspection mechanisms for Gen 6 BMW parts across Bavarian plants
- Engineered 3D computational geometry and 2D image-processing pipelines

Feb 2024 – Nov 2024 **Sono Motors** München, Germany
Simulation and Data Modeling Engineer (Working Student)
- Lead the implementation a solar energy yield simulation suite from the ground up
- Built solar-yield simulation suite, improving pipeline speed by 10×
- Designed and implemented an automated system health check framework for telematics data, decreasing average troubleshooting time by 90%

June 2022 – June 2023 **Ubicro Technology** Istanbul, Turkey
Chief Technology Officer
- Lead the software and mechanical design of an automated aeroponic agriculture system in our startup
- Finalized the mechanical design of the product and the manufacturing pipeline for plastic injection

Feb 2021 – Dec 2021 **General Electric Aviation** Istanbul, Turkey
Software Engineer (Part-Time)
- Conducted an in-depth evaluation of several open source Finite Element Analysis software, resulting in a recommendation that streamlined modeling processes and reduced simulation time by 50%
- Implemented automated simulation and analysis pipelines.

July 2020 – Feb 2021 **Siemens** Istanbul, Turkey
Software Engineer (Part-Time)
- Built and maintaned edge computing mechanisms for CNC machining
- Consulted on the Python API for the users and increased the efficiency of the deployment pipeline

Publications

- O. Subasi, A. Oral, S. Torabnia, **D. Erdogan**, M. B. Erdogan, I. Lazoglu. "In Silico Analysis of Elastomer-Coated Cerclage for Reducing Sternal Cut-Through in High-Risk Patients". Journal of Biomechanical Engineering. DOI: 10.1115/1.4050912

Awards

- Best Thesis Award (Mechanical Engineering)
- Koç University Academic Scholarship
- Highest ranking student award in Mechanical Engineering
- Dean's Academic Excellence Award (×5)

Skills

- High-Performance and Scientific Computing
- Numerical Modeling and Simulation. Finite Element Analysis. (Siemens NX, ANSYS, CAD, 3D Printing)
- Machine Learning, Deep Learning, Computer Vision
- Python, C/C++, OpenMP, MPI, MATLAB, Unix, Docker, LaTeX, SQL
- GRE Quantitative: 168, Verbal: 154

Languages

- English (C1)
- Turkish (Native)
- German (A2; B1.1 enrolled)

Volunteer Experience

Summer 2019 **AIESEC Orphanage, Cairo** Cairo, Egypt

Humanitarian Volunteer

- Worked at an orphanage as a full time volunteer for 8 weeks in Cairo, Egypt in summer of 2019. This was an AIESEC project mostly focusing on helping disabled people. We would teach the kids English and computer skills along with completing their daily exercises with them.